

Ancestry instrumentation for `fmix`

Why the origin-based transport meters stopped working,
what replaced them, and what the replacement measured

28 July 2026

Figures: `reports/ancestry_20260728/`.

1 The meters that broke

`fmix` has carried three positional-mixing meters since the directional-walk work: `odiff` (per-origin-family spread), `oadj` (adjacent-origin autocorrelation) and `disp` (mean normalised displacement). All three read the per-gate `origin` tag, the index of the input gate a gate descends from.

On the restructured menu they all read “nothing has moved”: `odiff` = 0.0001, `oadj` = 1.0000 after 40M moves and $2.58\times$ growth. Taken at face value that says the circuit is still, in ancestry layout, the original gate order.

It does not say that. A DB splice over a window spanning *mixed lineage* stamps its products `ORIGIN_SYNTH`, and all three statistics *skip* such gates. They are therefore computed over the material mixing has *failed to touch*, and they grow more selective the better the mixing works—survivorship bias pointing exactly the wrong way.

`osyn` was added to make the bias visible rather than arguable. It is the fraction of gates whose ancestry label has been destroyed:

moves	250k	1M	1.75M	2.5M	3M
<code>osyn</code>	22.9%	50.7%	63.3%	72.3%	77.0%

Half the circuit by 1M moves; on the small-circuit runs below it reaches 1.000, so there the three origin meters are not degraded but *undefined*.

Rule. Read `osyn` first. Once it is high, the origin meters carry no information.

2 What replaced them

A set union has no such failure mode: merging two lineages *adds* information where a scalar label must discard it.

Each *litter*—the set of gates one replacement emitted—carries the set of original input gates that contributed to it: the union of the sets of the litters its outgoing window drew on. Splits already propagate the parent’s litter id and so inherit the set for free, which is why this hooks four places rather than the thirteen that assign gate metadata: a splice unions its outgoing window; a merge unions both parents (ancestry comes from both, though provenance picks a side); born-random material has none; and input gate i is litter i .

Singleton sets are not stored—any litter id below the input count with no map entry denotes $\{id\}$, making initialisation $O(1)$ rather than $O(\text{input}^2)$ —and sets are pruned to live litters at each report. Cost is $|\text{input}|$ bits per live litter, so the instrument refuses above 20,000 input gates and is small-input by construction. All runs below use a 20,000- or 10,000-gate prefix of an $n = 16$ Gray-fold gadget (64 wires).

meter	per	units	question
anc	current gate	input gates	how many originals does it descend from?
span	current gate	input positions	how far apart were they?
fanout	<i>input</i> gate	current gates	how many gates carry this input?

Two corrections made along the way. A *normalised* span was the wrong measure: reporting span as a fraction of the circuit implies 1.0 is the target—that information should reach across the whole circuit—which was never the goal. Span is reported in absolute input-gate positions.

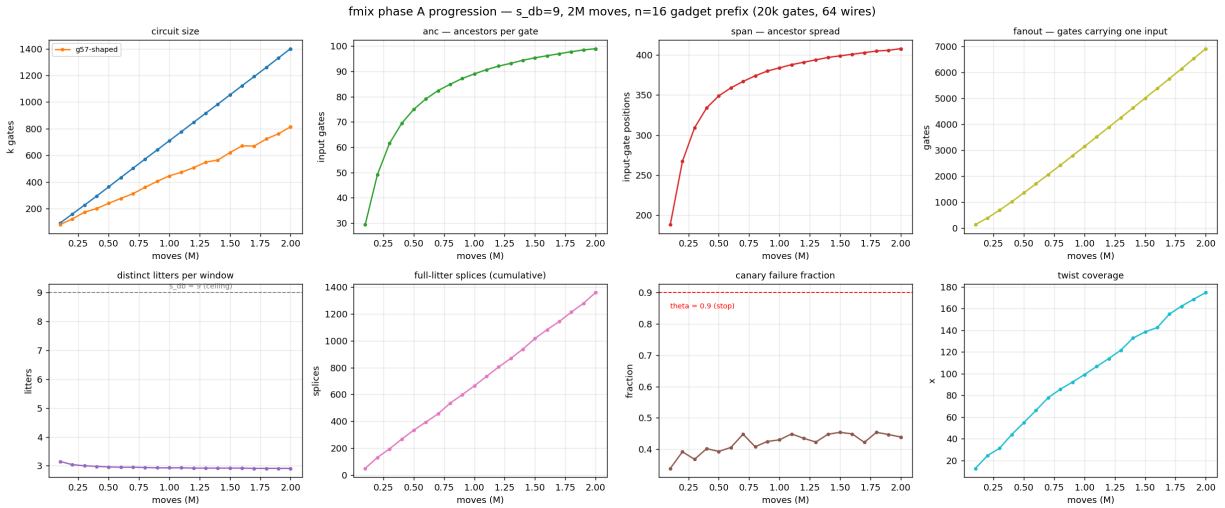
And **fanout** is redundant given **anc** and size, since both sum the same incidence relation:

$$\text{mean_fanout} = \text{mean_anc} \times \frac{\text{gates}}{\text{inputs}}.$$

At $s_{db} = 9$: $69.4 \times 293,341 / 20,000 = 1018$ against 1015 reported; at 10: 904 against 900. A fan-out *turnover* can therefore be a size effect wearing a spreading costume—and in the sweep below, it was. **span is the one carrying independent information**, describing the geometry of the ancestor sets rather than their size, and bounding how well an adversary can localise which part of the original computation a gate came from.

3 Results

3.1 Progression: 2M moves at $s_{db} = 9$

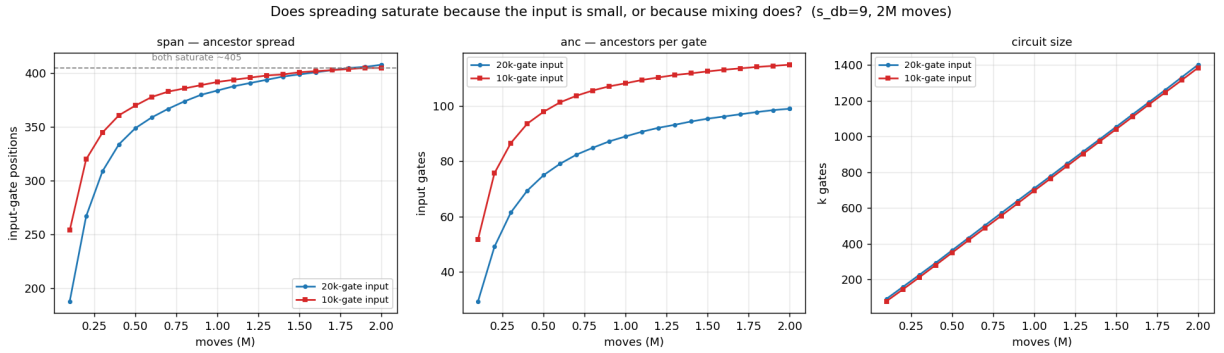


	start	2M
size	92k	1,401k
anc	29.4	99.0
span	188	408
fanout	135	6,909
distinct litters/window	3.15	2.91
canary	0.339	0.439
osyn	0.983	1.000

Ancestry climbs throughout but decelerates: Δanc per 100k falls from +32 to +0.5, Δspan from +121 to +2, both approaching an asymptote ($\text{anc} \approx 105$, $\text{span} \approx 415$).

Litter diversity is flat and low—3.15 $\rightarrow$ 2.91 against a ceiling of 9—and full-litter splices accrue at a dead-constant ~ 70 per 100k moves from the first interval to the last. That is an equilibrium reached almost immediately, not a transient: windows keep being drawn from ~ 3 replacement events even though they span 9 gates.

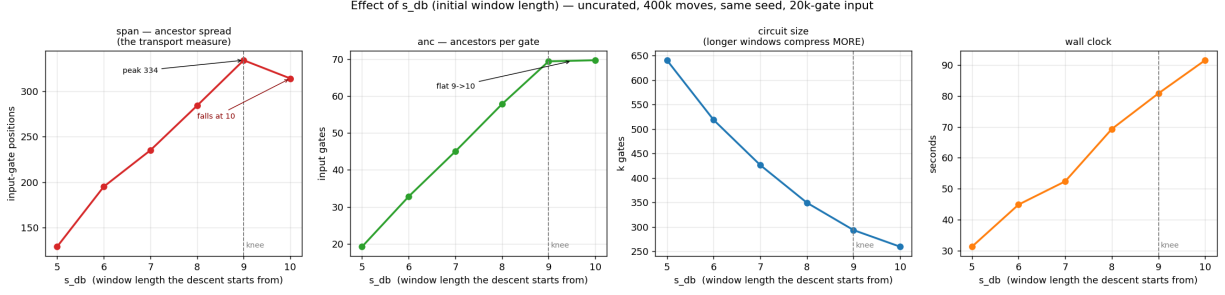
3.2 Is saturation the input’s fault? 10k versus 20k



	at 2M moves	20k input	10k input
span		408	405
anc		99.0	114.9
size		1,401k	1,384k
growth ratio		70 $\times$	138$\times$

Span lands on the same value from a halved input, across runs differing in both input size and growth ratio. Saturation is a property of the *mixing*, not a ceiling from the original circuit. **anc** being higher on the smaller input fits: same span in absolute positions, but 138 $\times$ versus 70 $\times$ growth means more splices per input gate. Growth drives **anc**; something else pins **span**.

3.3 What pins span: the s_{db} sweep



Uncured, 400k moves, same seed, 20k input.

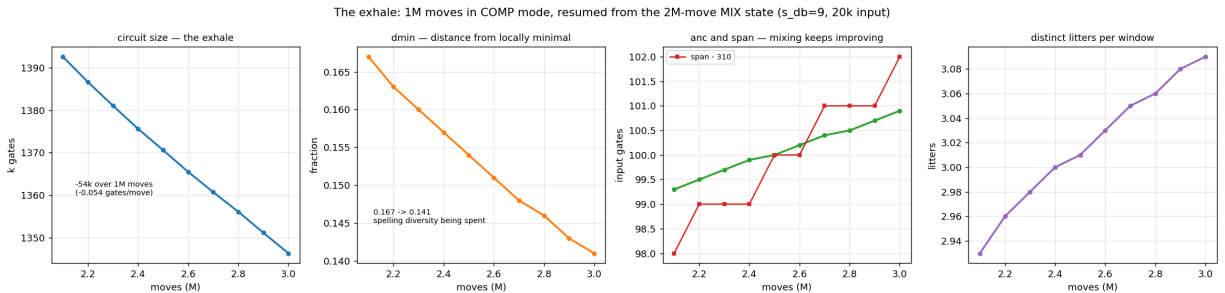
s_{db}	anc	span	fanout	size	wall
5	19.2	129	614	640,596	31.3 s
6	32.8	195	848	518,594	44.9 s
7	45.0	235	956	426,263	52.4 s
8	57.8	284	1006	349,262	69.3 s
9	69.4	334	1015	293,341	80.8 s
10	69.7	314	900	259,335	91.5 s

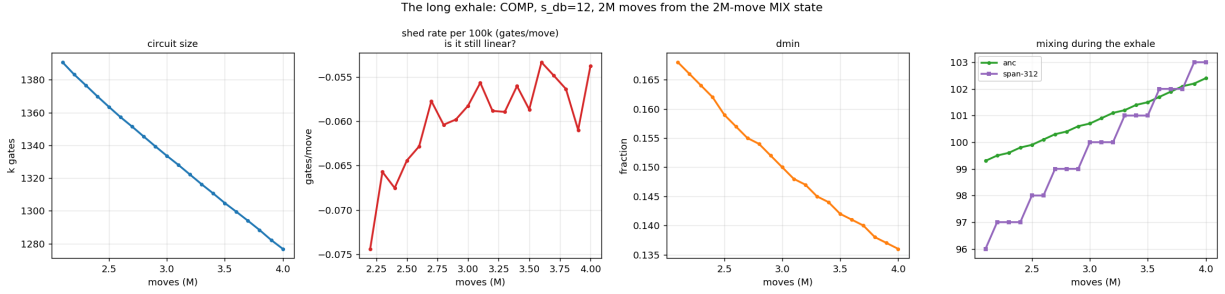
Span rises $\sim +51$ positions per extra gate of window, then turns at 10; **anc** goes flat exactly where span turns ($69.4 \rightarrow 69.7$)—the same number of ancestors, packed tighter. So span is a *tunable*, not a limit: the reach of the mixing operator is set by how much a window spans, and information travels only as far as overlapping windows carry it. Running longer buys **anc**; it buys almost no **span** after the first ~ 500 k moves.

Size falls 60% across the sweep and keeps falling past the knee. Longer windows find free (non-growing) matches far more often—the splice histogram gives the free fraction as 0/10.5/23.1/39.9/68.8% at lengths 1–5—so they pay less. Past the knee that extra length is spent on *compression* rather than combination: the 10-rung behaves like a COMP move embedded in MIX, pulling toward local minimality. A built-in damper.

Recommendation. s_{db} of 8–9 rather than the spec’s 5: $3.6\times$ the **anc**, $2.6\times$ the span, and a circuit $2.2\times$ smaller, for $2.6\times$ the wall clock. The knee should be re-measured on the $n = 128$ gadget, since the store’s decay with window length is material-dependent.

3.4 The exhale: COMP mode





COMP, $s_{db} = 12$, 2M moves	start	end
size	1,400,600	1,276,860
shed rate	-0.074	-0.056 (flat)
dmin	0.168	0.136
anc	99.3	102.4
span	408	415
distinct litters	2.97	3.54

COMP wants a longer window than MIX: -0.060 gates/move at $s_{db} = 12$ against -0.054 at 9, consistent with the sweep, where size kept falling past the mixing knee.

Mixing does *not* degrade during an exhale. **anc** and **span** both rise, and litter diversity makes its largest move of any run. COMP-DB splices are still re-encodings—they combine ancestry and stamp generations—so compression is not idle time.

The cost is entirely in **dmin**, which falls steadily ($0.168 \rightarrow 0.136$, about 19% of the way to zero) without decelerating: spelling diversity being spent, the circuit moving toward the locally-minimal form that **fcompress**, being attacker-computable, would reach anyway.

No bend appears in 2M moves of exhale—the shed rate decays $\sim 20\%$ early then goes flat—so by the “bend marks the natural inhale point” criterion there is a long way still to go.

4 What this changes

1. **odiff**, **oadj** and **disp** should not be quoted without **osyn** beside them, and not at all once it is high. Earlier transport conclusions drawn from them—including “transport is exactly zero after $2.58\times$ growth”—are unsupported.
2. **span** is the transport measure; **fanout** is **anc** times the size ratio and adds nothing; normalised span encodes a target we do not have.
3. Span saturates for reasons internal to the mixing, and the lever is window length, not run length.
4. s_{db} should be larger than the spec’s 5, with the knee at 8–9 on this material, and COMP plausibly wanting more than MIX.
5. Litter diversity is the untouched lever: windows draw from ~ 3 replacement events against a ceiling of s_{db} , at equilibrium from the start.

5 Open

- Re-measure the s_{db} knee on the $n = 128$ gadget (689k gates).
- Locate the shed-rate bend with a long exhale, and test whether the other statistics turn at the same point.
- `-litter-samples` and `-litter-ban` are built and untested against these baselines.
- Contiguous versus convex sampling has never been separated ($p_{\text{convex}} = 0.5$ throughout); COMP and MIX may want different geometry.
- `p_mingen` has been 0.8 throughout, including in COMP where weaker pool targeting may serve better.
- What counts as “enough” span needs a reference scale from the construction, not from the circuit’s length.